Building a Family Genealogical Chart

Practical Assignment · Genetics & Heredity

Goal: Construct a family pedigree chart using standard genetic symbols, strict generation levels, and phenotype data collected from your relatives.

Symbols Key

	Square	Male
	Circle	Female
	Crossed-out shape	Deceased relative
	Solid black shape	Individual displaying the trait/condition being tracked
	Horizontal line	Marriage / partnership — connects a male and a female
	Vertical line + sibship line	Drops from the marriage line to the children

Instructions

1. Organize by generation levels

- Representatives of the same generation (peers) must sit **strictly on the same horizontal line**.
- **Level I (bottom):** you, your siblings, and first cousins.
- **Level II:** parents, aunts, and uncles.
- **Level III:** grandparents.
- **Level IV (top):** great-grandparents.
- Connect spouses with a horizontal line, then draw a vertical line down to their children, arranged **left to right from oldest to youngest**.
- People who married into the family (e.g. an uncle's wife) sit on the same level as their spouse.

2. Data collection — interview relatives and record for each person:

- Eye color · Hair color · Height (cm)
- Current age (living relatives) or age at death
- Cause of death (if known)

Be respectful: some relatives may prefer not to share certain details. Write "unknown" or "not shared" — that is perfectly acceptable.

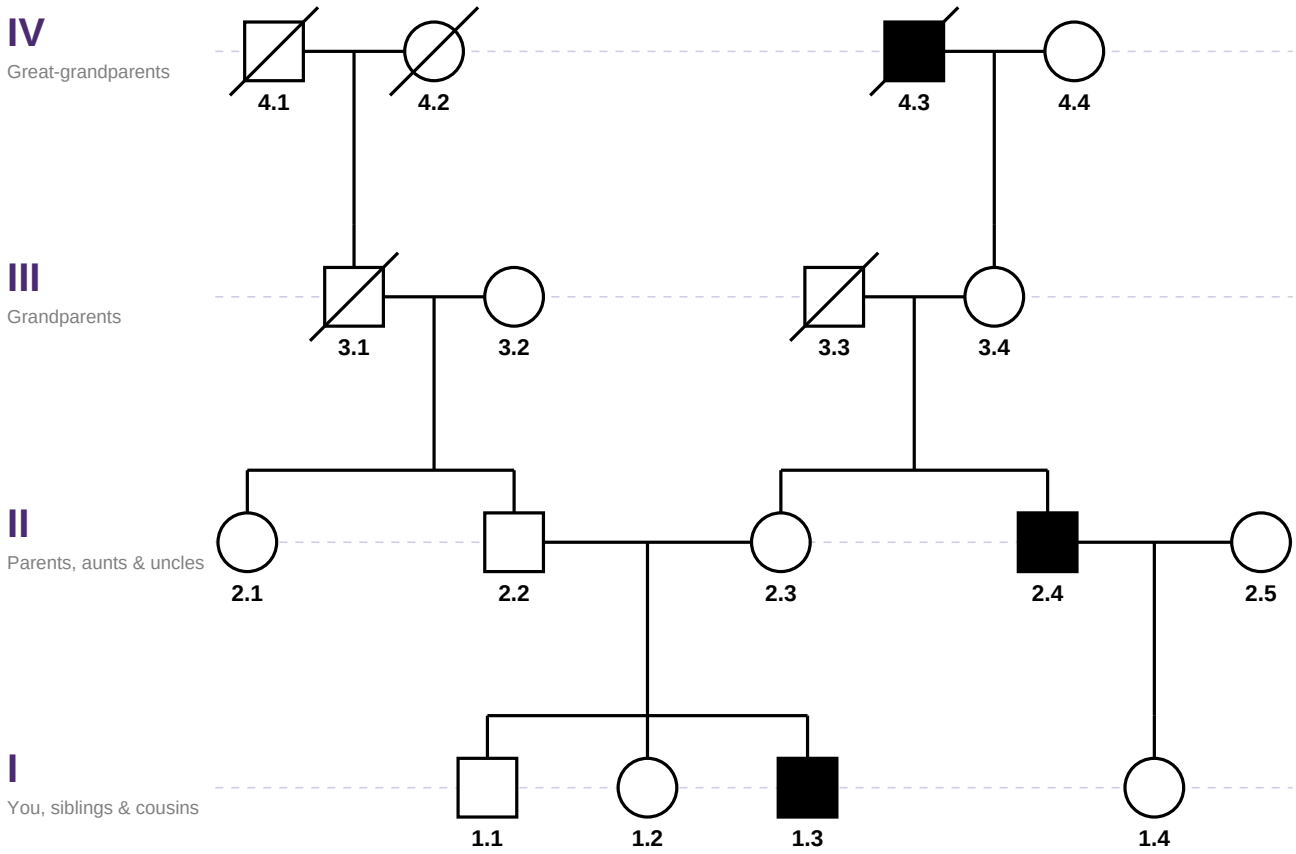
Pro Tip Keep your chart clean: give every shape a number — **1.1, 1.2…** for Level I, **2.1, 2.2…** for Level II, and so on (numbering left to right). Write the trait details on a **separate sheet** as a numbered list or table.

Submission Format

- A graphical chart (A4 / A3 paper or digital)
- A separate numbered trait index (list or table — template on page 3)

Worked Example

A small, correctly drawn example. Notice that each generation sits on one line, siblings hang from a shared sibship line, and people who married in (2.5) have no parents drawn.



Here the tracked trait appears in 4.3 → 2.4 and 1.3 (solid shapes).

Example trait index (extract)

No.	Name	Relation	Sex	Eyes	Hair	Height	Age	Cause of death
1.2	Ana	Me	F	green	blonde	162 cm	15	—
1.3	Luka	Brother	M	brown	dark brown	168 cm	13	—
2.4	Marko	Uncle (mother's side)	M	blue	brown	181 cm	44	—
3.1	Janez	Grandfather (father's side)	M	brown	black	176 cm	d. 71	heart attack
4.4	Marija	Great-grandmother	F	grey	grey	155 cm	94	—

Trait Index — Template

Fill one row per person on your chart. Number them exactly as on the chart.

No.	Name	Relationship to you	Sex	Eye color	Hair color	Height (cm)	Age / age at death	Cause of death (if known)
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